CLOSED IMMERSIONS IDENTIFY QUOT FUNCTORS BY THE PI_* EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 900-908 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 910-922, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 25 of queue_moduli.json (pb=620, stmt 900-908, proof 910-922).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DP1: sources/stacks-moduli/pinned-original-source.tex, source lines 828–834. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04CJ: sources/stacks-moduli/pinned-original-source.tex, source lines 922–922. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let B be an algebraic space. Let $\pi : X \to Y$ be a closed immersion of algebraic spaces which are separated and of finite presentation over B. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then the morphism* $\mathrm{Quot}_{\mathcal{F}/X/B} \to \mathrm{Quot}_{\pi_*\mathcal{F}/Y/B}$ *of Lemma 0DP1 is an isomorphism.*

Proof. For every scheme T over B the morphism $\pi_T : X_T \to Y_T$ is a closed immersion. Then $\pi_{T,*}$ is an equivalence of categories between $QCoh(\mathcal{O}_{X_T})$ and the full subcategory of $QCoh(\mathcal{O}_{Y_T})$ whose objects are those quasi-coherent modules annihilated by the ideal sheaf of X_T, see Morphisms of Spaces, Lemma 04CJ. Since a qotient of $(\pi_*\mathcal{F})_T$ is annihilated by this ideal we obtain the bijectivity of the map $\mathrm{Quot}_{\mathcal{F}/X/B}(T) \to \mathrm{Quot}_{\pi_*\mathcal{F}/Y/B}(T)$ for all T as desired. □

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.